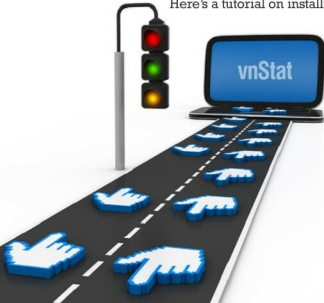


vnStat: A Lightweight Network Traffic Monitor

vnStat is a Web based interface to view and monitor network traffic bandwidth usage in a nice graphical format. It is a command line tool and its integration with PHP gives it a graphical frontend. Here's a tutorial on installing and using vnStat with a PHP frontend.



There are occasions when it is useful to know your bandwidth usage pattern. This not only helps administrators to detect the root cause of traffic related issues, such as a network overload, but also helps them to keep a tab on traffic flow to and from the Internet, and the charges levied by the ISP for the bandwidth used.

This is an important task which requires dedicated and effective software. vnStat is just the tool for that. It is a lightweight (command line) network traffic monitor, which provides several output options like summaries on an hourly, daily, monthly, weekly or 'top 10 days' basis. It can monitor selected interfaces, stores network traffic logs in a database for further analysis and can be used without root permission.

Prerequisites for installation

I have chosen a system with CentOS version 6.7 (equivalent to RHEL 6.7) with minimal installation, which acts as a gateway through which eth0 is connected to the Internet and eth1 is connected to the local network. First, install all the required packages and dependencies for vnStat and the PHP Web frontend using the Yum utility.

```
[root@sun ~]# yum install wget epel-release php httpd php-gd
```

Installation

To install vnStat, execute the command given below in a terminal:

```
[root@sun ~]# yum install vnstat
```

To view all the available interfaces on your system that vnStat can monitor, use the following command:

```
[root@sun ~]# vnstat --iflist
Available interfaces: lo eth0 eth1
```

Now add the eth0 and eth1 interfaces for monitoring traffic, as follows:

```
[root@sun ~]# vnstat -u -i eth0
Error: Unable to read database "/var/lib/vnstat/eth0"
Info: -> A new database has been created
```

For the first time, it will display the above error. Ignore this as sufficient data is not available for display.

Similarly, add the second interface, i.e., eth1, as follows:

```
[root@sun ~]# vnstat -u -i eth1
Error: Unable to read database "/var/lib/vnstat/eth1".
Info: -> A new database has been created.
```

Method for updating the database

The database can be updated by either using cron or service.

Cron: To use cron, just remove the comment from */etc/cron.d/vnstat*, after which the entry should be as shown below:

```
#vi /etc/cron.d/vnstat
MAILTO=root
*/5 * * * * vnstat /usr/sbin/vnstat.cron
```

:wq save and quit

Service: The second method uses vnStatd a program that collects all the information and is run as the daemon. You can start it with executing below command:

```
[root@sun ~]# /etc/init.d/vnstat start
```

Enable service at boot time

```
[root@sun ~]# chkconfig --level 345 vnstat on
```

To start traffic monitoring for selected interfaces, start vnstat service.

```
[root@sun ~]# cd /usr/local/src
```

Download the latest vnStat PHP frontend